

### (a) Twin construction



$A$ : dog     $B$ : rooster

$q_+$ : dog followed by rooster  $\rightarrow W_+$

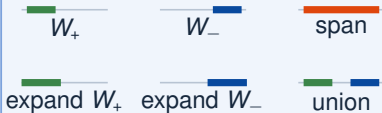
$q_-$ : rooster followed by dog  $\rightarrow W_-$

### (b) RelTwin training

#### Stage 1: Answer learning

|       | $y(W_+)$            | $y(W_-)$            |
|-------|---------------------|---------------------|
| $q_+$ | $s_{++} \uparrow$   | $s_{+-} \downarrow$ |
| $q_-$ | $s_{-+} \downarrow$ | $s_{--} \uparrow$   |

#### Stage 2: Set preferences



### (c) Window selection

